# Splitting the $L_{1/2}$&TV Problem into Three Subproblems

Compound-regularized linear inverse problem [1], with echo $\mathbf{y} = \mathrm{vec}(\mathbf{Y})$, scene $\mathbf{h} = \mathrm{vec}(\mathbf{H})$ ($\mathbf{Y}, \mathbf{H} \in \mathbb{C}^{M\times N}$, $M$ azimuth and $N$ range samples; vec stacks the columns), CSA matrix $\mathbf{\Phi}$ with $\mathbf{h}_{\mathrm{CSA}} = \mathbf{\Phi}\mathbf{y}$, so the echo model is $\mathbf{\Phi}^{-1}$; total variation $\mathrm{TV}(\cdot)$ [2] and penalty weights $\lambda_1, \lambda_2 > 0$. Introducing auxiliary variables [3] and then the Lagrangian with multipliers $\xi_1, \xi_2 > 0$ (kept fixed):

$$\hat{\mathbf{h}} = \arg\min_{\mathbf{h}} \left\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\right\|_2^2 + \lambda_1\|\mathbf{h}\|_{1/2} + \lambda_2\,\mathrm{TV}(\mathbf{h}) \tag{1}$$

$$= \arg\min_{\mathbf{h},\mathbf{z}_1,\mathbf{z}_2} \left\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\right\|_2^2 + \lambda_1\|\mathbf{z}_1\|_{1/2} + \lambda_2\,\mathrm{TV}(\mathbf{z}_2) \quad \text{s.t.} \quad \|\mathbf{h}-\mathbf{z}_1\|_2^2 = 0, \quad \|\mathbf{h}-\mathbf{z}_2\|_2^2 = 0 \tag{2}$$

(introduce auxiliary variables $\mathbf{z}_1, \mathbf{z}_2$: $\|\mathbf{h}-\mathbf{z}\|_2^2 = 0 \iff \mathbf{z} = \mathbf{h}$)

$$\approx \arg\min_{\mathbf{h},\mathbf{z}_1,\mathbf{z}_2} \underbrace{\left\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\right\|_2^2 + \lambda_1\|\mathbf{z}_1\|_{1/2} + \lambda_2\,\mathrm{TV}(\mathbf{z}_2) + \xi_1\|\mathbf{h}-\mathbf{z}_1\|_2^2 + \xi_2\|\mathbf{h}-\mathbf{z}_2\|_2^2}_{\Gamma(\mathbf{h},\mathbf{z}_1,\mathbf{z}_2)} \tag{3}$$

(Lagrangian: move the constraints into the objective)

The minimization over $(\mathbf{h}, \mathbf{z}_1, \mathbf{z}_2)$ returns its $\mathbf{h}$-component. With $\xi_1, \xi_2$ fixed, $\xi\|\mathbf{h}-\mathbf{z}\|_2^2$ penalizes a copy that drifts away from $\mathbf{h}$ [3], so (3) approximates (2).

Alternating minimization of $\Gamma$ (one variable at a time, the others at their latest values; terms without that variable are constants) splits (3) into three subproblems:

$$\mathbf{h}^{(k+1)} = \arg\min_{\mathbf{h}} \ \Gamma\big(\mathbf{h}, \mathbf{z}_1^{(k)}, \mathbf{z}_2^{(k)}\big) = \arg\min_{\mathbf{h}} \ \left\|\mathbf{y} - \mathbf{\Phi}^{-1}\mathbf{h}\right\|_2^2 + \xi_1\left\|\mathbf{h} - \mathbf{z}_1^{(k)}\right\|_2^2 + \xi_2\left\|\mathbf{h} - \mathbf{z}_2^{(k)}\right\|_2^2, \tag{4}$$

$$\mathbf{z}_1^{(k+1)} = \arg\min_{\mathbf{z}_1} \ \Gamma\big(\mathbf{h}^{(k+1)}, \mathbf{z}_1, \mathbf{z}_2^{(k)}\big) = \arg\min_{\mathbf{z}_1} \ \xi_1\left\|\mathbf{z}_1 - \mathbf{h}^{(k+1)}\right\|_2^2 + \lambda_1\|\mathbf{z}_1\|_{1/2}, \tag{5}$$

$$\mathbf{z}_2^{(k+1)} = \arg\min_{\mathbf{z}_2} \ \Gamma\big(\mathbf{h}^{(k+1)}, \mathbf{z}_1^{(k+1)}, \mathbf{z}_2\big) = \arg\min_{\mathbf{z}_2} \ \xi_2\left\|\mathbf{z}_2 - \mathbf{h}^{(k+1)}\right\|_2^2 + \lambda_2\,\mathrm{TV}(\mathbf{z}_2). \tag{6}$$

- (4): quadratic in $\mathbf{h}$, solved in closed form below.
- (5): $L_{1/2}$ denoising of $\mathbf{h}^{(k+1)}$, solved entrywise by half thresholding [4].
- (6): TV denoising of $\mathbf{h}^{(k+1)}$, solved by Chambolle's projection algorithm [5].

In practice $\xi_1 = \xi_2 = \xi$. Dividing by $\xi$, (5) minimizes $\|\mathbf{z}_1 - \mathbf{h}^{(k+1)}\|_2^2 + (\lambda_1/\xi)\|\mathbf{z}_1\|_{1/2}$ and (6) minimizes $\frac{1}{2}\|\mathbf{z}_2 - \mathbf{h}^{(k+1)}\|_2^2 + \big(\lambda_2/(2\xi)\big)\,\mathrm{TV}(\mathbf{z}_2)$, so the effective weights $\lambda_1/\xi$ and $\lambda_2/(2\xi)$ are tuned through $\lambda_1, \lambda_2$.

**Subproblem 1: image update.** $\boldsymbol{\Phi}$ is unitary ($\boldsymbol{\Phi}^{-1} = \boldsymbol{\Phi}^H$), so $\|\mathbf{y} - \boldsymbol{\Phi}^{-1}\mathbf{h}\|_2^2 = \|\boldsymbol{\Phi}^H(\mathbf{h}_{\text{CSA}} - \mathbf{h})\|_2^2 = \|\mathbf{h}_{\text{CSA}} - \mathbf{h}\|_2^2$. Writing each squared norm as a sum over the $MN$ entries, with $[v_n]_{n=1}^{MN}$ the vector with entries $v_n$:

$$
\begin{aligned}
\mathbf{h}^{(k+1)} &= \arg\min_{\mathbf{h}} \ \big\|\mathbf{y} - \boldsymbol{\Phi}^{-1}\mathbf{h}\big\|_2^2 + \xi_1\big\|\mathbf{h} - \mathbf{z}_1^{(k)}\big\|_2^2 + \xi_2\big\|\mathbf{h} - \mathbf{z}_2^{(k)}\big\|_2^2 \\
&= \arg\min_{\mathbf{h}} \ \big\|\mathbf{h}_{\text{CSA}} - \mathbf{h}\big\|_2^2 + \xi_1\big\|\mathbf{h} - \mathbf{z}_1^{(k)}\big\|_2^2 + \xi_2\big\|\mathbf{h} - \mathbf{z}_2^{(k)}\big\|_2^2 \\
&\qquad (\boldsymbol{\Phi} \text{ unitary}) \\
&= \arg\min_{\mathbf{h}} \ \sum_{n=1}^{MN}\Big[\,\big|h_{\text{CSA},n} - h_n\big|^2 + \xi_1\big|h_n - z_{1,n}^{(k)}\big|^2 + \xi_2\big|h_n - z_{2,n}^{(k)}\big|^2\,\Big] \\
&\qquad (\|\mathbf{v}\|_2^2 = \sum_n |v_n|^2) \\
&= \arg\min_{\mathbf{h}} \ \sum_{n=1}^{MN}\Big[\,\big(\mathrm{Re}\{h_n\} - \mathrm{Re}\{h_{\text{CSA},n}\}\big)^2 + \big(\mathrm{Im}\{h_n\} - \mathrm{Im}\{h_{\text{CSA},n}\}\big)^2 \\
&\qquad\qquad + \xi_1\big(\mathrm{Re}\{h_n\} - \mathrm{Re}\{z_{1,n}^{(k)}\}\big)^2 + \xi_1\big(\mathrm{Im}\{h_n\} - \mathrm{Im}\{z_{1,n}^{(k)}\}\big)^2 \\
&\qquad\qquad + \xi_2\big(\mathrm{Re}\{h_n\} - \mathrm{Re}\{z_{2,n}^{(k)}\}\big)^2 + \xi_2\big(\mathrm{Im}\{h_n\} - \mathrm{Im}\{z_{2,n}^{(k)}\}\big)^2\,\Big] \\
&\qquad (|u - h_n|^2 = (\mathrm{Re}\{h_n\} - \mathrm{Re}\{u\})^2 + (\mathrm{Im}\{h_n\} - \mathrm{Im}\{u\})^2) \\
&= \frac{1}{1+\xi_1+\xi_2}\Big[\,\mathrm{Re}\big\{h_{\text{CSA},n} + \xi_1 z_{1,n}^{(k)} + \xi_2 z_{2,n}^{(k)}\big\} + j\,\mathrm{Im}\big\{h_{\text{CSA},n} + \xi_1 z_{1,n}^{(k)} + \xi_2 z_{2,n}^{(k)}\big\}\,\Big]_{n=1}^{MN} \\
&\qquad \text{(the sum separates into } 2MN \text{ one-dimensional quadratics, one in each } \mathrm{Re}\{h_n\} \text{ and } \mathrm{Im}\{h_n\}; \\
&\qquad\ \text{minimize each by setting its derivative to zero)} \\
&= \frac{\mathbf{h}_{\text{CSA}} + \xi_1\mathbf{z}_1^{(k)} + \xi_2\mathbf{z}_2^{(k)}}{1+\xi_1+\xi_2}. \qquad (7) \\
&\qquad (\mathrm{Re}\{c\} + j\,\mathrm{Im}\{c\} = c \text{ for each entry})
\end{aligned}
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## References


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